Project Report: Computer Vision-Based Chair Sit and Reach Flexibility Test

1. Introduction

Flexibility is a critical component of physical fitness and is commonly measured using the sit-and-reach test. Traditional measurement methods require manual assessment using a ruler, which may be inconvenient or inaccurate without another person's help.

This project introduces a computer vision–based system that uses Mediapipe Pose and OpenCV to automate the measurement process, providing real-time guidance and feedback during the flexibility test.

2. Objectives

- Develop a state-machine–based system to guide users step by step through the flexibility test.
- Ensure real-time posture validation (e.g., sitting position, foot flat, leg extended, hand placement).
- Automate measurement of fingertip-to-toe distance in normalized 3D space.
- Provide instant feedback (Above Average / Average / Below Average flexibility).
- Create a user-friendly interface for testing and debugging.

3. Tools and Technologies

- Programming Language: Python
- Libraries:
  - mediapipe (Pose detection and landmark tracking)
  - opencv-python (Video capture, display, visualization)
  - numpy (Mathematical calculations)
  - math (Angle and distance computations)
- Hardware: Standard webcam (inexpensive webcam tested)

4. System Design

Workflow:
1. Camera Input: Video captured from webcam.
2. Pose Detection: Mediapipe detects 3D body landmarks.
3. State Machine: Guides user through stages (Full-body detection → Sitting → Foot flat → Leg extended → Hands stacked → Inhale/Exhale → Reach → Hold → Measure → Complete)
4. Flexibility Measurement: Distance calculated between fingertips and extended foot.
5. Feedback Displayed: "Above Average", "Average", or "Below Average".

5. Implementation

Key Functions:
- Pose Validation: check_sitting_position(), check_foot_flat_on_floor(), check_leg_extended(), check_hands_position(), check_forward_reach()
- Measurement: measure_flexibility() → returns distance and feedback.
- Debugging Tool: debug_print_landmarks() prints all landmark coordinates.

User Interaction:
- System shows real-time instructions on screen.
- Holds each stage for instruction + 2 seconds validation.
- Final measurement result displayed until user presses any key.

6. Results

- Successfully guided users through flexibility test.
- Accurate detection of sitting posture, foot placement, and hand alignment.
- Measurement results provided in inches with feedback (Above/Below Average).
- Tested with low-resolution webcam, system performed reliably with tuned thresholds.

7. Limitations & Future Work

- Depth accuracy limited by 2D camera and Mediapipe's estimated z-values.
- Currently supports only sit-and-reach test; other flexibility tests can be added.
- Future improvements:
  - Multi-camera setup for better 3D accuracy.
  - Mobile app integration.
  - AI-based scoring adjusted for age and gender.
  - Online Website

8. Conclusion

This project demonstrates a low-cost, computer vision–based solution for flexibility assessment using Mediapipe and OpenCV. The automated sit-and-reach test reduces reliance on other people, manual measurement tools and provides real-time, interactive guidance and feedback.

It has applications in senior fitness, sports, and rehabilitation, making flexibility testing more accessible and accurate.

9. References

- Mediapipe Documentation: https://developers.google.com/mediapipe
- OpenCV Documentation: https://opencv.org
- Fitness Research: Sit-and-Reach Test protocols